

Assignment

Module 1.3 RAM and Storage

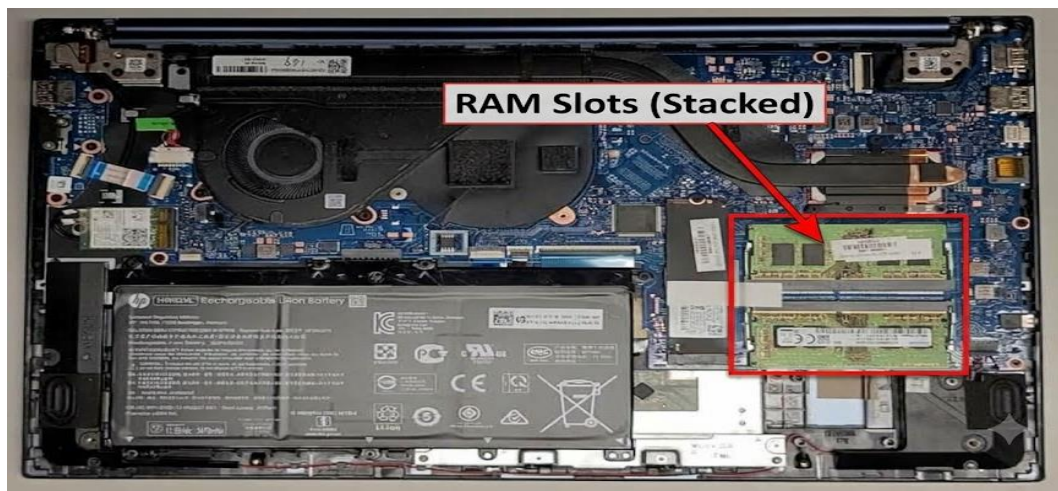
1. Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.



Technical Parameter	DDR3	DDR4	DDR5
Memory Generation	3rd-generation DDR	4th-generation DDR	5th-generation DDR
Typical Data Speed	800–2133 MT/s	1600–3200 MT/s	4800–6400+ MT/s
Operating Voltage	~1.5 V	~1.2 V	~1.1 V
Memory Architecture	64-bit DIMM	64-bit DIMM	64-bit DIMM with two 32-bit channels
Power Management	Motherboard-based	Motherboard-based	On-DIMM PMIC
Memory Capacity	Lower	Higher	Significantly higher
Bandwidth	Low	Improved	Very high
Typical Uses	Older PCs, laptops, servers	Mainstream PCs and servers	Modern PCs, gaming, workstations, servers

2. Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully.

Hint: Refer to your device manual or a trusted YouTube guide if unsure about RAM removal and installation steps.



3. **List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.**



HDD: A hard disk drive is an older storage option that uses spinning disks with a magnetic coating to keep our data. It works slower than an SSD, but it costs less money and can hold much more of our information. For example, we can use an HDD to keep our family photos, videos, files, and backup copies, since these items take up lots of space, but we don't need to access them right away.

SSD: A solid-state drive saves data using flash memory and has no moving parts inside. It runs faster, makes less noise, and works better than an HDD for our normal computer tasks. For example, I can use an SSD for Gmail and web browsing because my computer starts up quickly and my programs and files open faster.

NVMe SSD: An NVMe SSD is a faster kind of SSD that uses the PCIe connection to move our data around. It gives us very fast speeds and short wait times. For example, I can use an NVMe SSD for Google Maps or other apps that I use regularly because my programs start up quickly and my system can access data with almost no wait.

To put it simply: HDD is best when we want more space for less money, SSD is good for our quick everyday computer work, and NVMe SSD is right when we need the fastest speed and best performance.

4. **Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly.

Constraint: Only perform this task if you have access to a device you are allowed to open. If not, find and share a YouTube video link showing the process for your laptop/PC model. **

- Here I am attaching a YouTube link which shows the whole process for removing my HP laptop storage drive.

https://youtu.be/5pUnyvfwfN38?si=NVaZvjQ_3qLKnUvW